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Rejection, Disagreement, Controversy and Acceptance in Mathematical Practice: Episodes in the Social Construction of Infinity

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Abstract

The concept of infinity has a long and troubled history. Thus it is a promising concept with which to explore rejection, disagreement, controversy and acceptance in mathematical practice. This paper briefly considers four cases from the history of infinity, drawing on social constructionism as the background social theory. The unit of analysis of social constructionism is conversation. This is the social mechanism whereby new mathematical claims are proposed, scrutinised and critiqued. Minimally, conversation is based on the two roles of proponent and critic. The proponent puts forward a proposal, which is reacted to and evaluated by those in the role of critic. There is a continuum of contexts in which such conversations take place from inner conversations the mathematician has within themselves, and casual face-to face interactions between mathematicians at the chalkboard, all the way to the formal responses of referees and editors to submitted journal papers. Such responses vary from unconditional acceptance, partial acceptance through to outright rejection. There may be disagreements between proponents and critics, among those in the joint role of critic, and broader, community-wide disagreements and controversies, according to specific mathematical proposal and the critical judgements of it.

Keywords Acceptance · Cantor · Conversation · Conversation theory · Critic · Dialectics · Dialogic · Disagreement · Infinity · Intuitionism · Proposer · Rejection · Social constructionism

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1 Introduction

The rich and complex concept of infinity has a long and troubled history, and it is probably the most controversial concept in mathematics. Thus it is a promising concept with which to explore rejection, disagreement, controversy and acceptance in mathematical practice. This paper briefly considers four cases from the history of infinity, the last two as yet largely undiscussed in the history and philosophy of mathematics.

2 Social Constructionism and Conversation

Rejection, disagreement, controversy, and acceptance are all social processes or outcomes, so a social theory of mathematics and mathematical practice is needed. In this paper I draw on social constructionism, which views mathematics as made up of different but overlapping social practices. Each social practice is an organised group activity that has an identity, members, agreed forms of participation, both implicit and explicit, and has above all, goals shared by all participants as the overarching focus. All aspects of a social practice can change over its lifetime, including its members, rules of active participation, overall structure, and even the goals. But when the goals change significantly it becomes a different social practice, for goals determine the whole point and direction of the practice (Ernest 2021).

A social practice in mathematics can encompass a university mathematics department, or it can be as wide as the community of mathematicians with shared or parallel research goals, such as a shared focus on a particular topic, problem, sub-specialism, or methodology. Members of a mathematical practice must be in communication either face to face or by other means, meeting in the same department, or at conferences, symposia, congresses. In the extreme case members of the same practice might never meet face to face and communicate exclusively via various distance forms of the media.

Lakatos (1976) is a seminal source of social constructionism in its emphasis on the historical development of mathematics. It shows how mathematical topics including the Euler relation and elements of analysis developed through the changing meanings and interpretations of key concepts, as well as modifications of proofs, in response to feedback and dialogical interactions. The growth and change followed a dialectical process of thesis-antithesis-synthesis. This the basis for the Logic of Mathematical Discovery of Lakatos (1976), extended in Ernest (1998). The dialectical process discussed here is an heuristic pattern, a loosely defined and suggestive argumentation structure closer to the dialectic of the Greeks. It is open to use and interpretation in a number of ways, and is not the ineluctable and rigid logic of dialectical materialism of Marxism/Leninism.

At the heart of social constructionism lies conversation theory (CT), with its dialogical pattern of interactions that lead to mathematical knowledge growth and

warranting. The basic manifestation of conversation takes place between two persons communicating within a shared social context. Conversation and dialogue have been widely used a fundamental epistemological notions, for example in Plato (Long 2013), Mead (1934), Wittgenstein (1953), Rorty (1979), Habermas' (1984) and Shotter (1991). Many other philosophers could be cited, including Bakhtin, Gadamer, Buber, Gergen and Harré. Within the philosophy of mathematics there is growing attention to conversational, dialogical and dialectical interpretations of mathematics (Ernest 1994, 1998; Dutilh Novaes 2021; Larvor 2001).

In its original form conversation is *inter*personal, consisting of persons exchanging speech, or other constellations of signs face-to-face. Two secondary forms of conversation are derived from this. First, there is *intra*personal conversation, i.e., thought as constituted and formed by conversation. According to CT (verbal) thinking is originally internalised conversation with an imagined other (Plato - see Long 2013; Vygotsky 1978; Mead 1934). Intrapersonal conversation becomes much more than 'words in the mind', and the conversational roles of proponent and critic become part of one's mental functions (Ernest and Sfard 2018). CT claims that most forms of creativity are based on or have a very important dimension of intrapersonal conversation, with the creator alternating between the roles of proponent and critic, most commonly supported by external material artifacts, such as proof notes, written text, or drawn figures.

A further form is cultural conversation, an extended variant of interpersonal conversation comprising the creation and exchange of texts, often in permanent (i.e., enduringly embodied) form. This includes chains of correspondence such as letters, papers, email messages, transmitted diagrams, etc., exchanged between persons. Such conversations can be extended over years, lifetimes, or even beyond, if new persons join in and maintain the conversation.

Conversation has an underlying dialogical form of ebb and flow, comprised of the alternation of voices in assertion and counter assertion. Conversation is the basis of all feedback, whether it be in the form of acceptance, elaboration, reaction, asking for reasons, correction, criticism or rejection. Such feedback is essential for all human knowledge growth and learning. In performing such functions the different conversational roles include the two main forms of proponent and critic, which occur in all of the modes but originate in the interpersonal.

The role of the proponent lies putting forward a reasoned argument, an emergent sequence of ideas, a line of thinking, a thought experiment or a narrative. The aim is to communicate an idea, an argument, to build understanding or to convince the listener. This is the role of the author of a mathematical text be it spoken, sketched or written. The proponent (author) constructs a knowledge representation and submits it to a publication outlet, be it formal or informal. In the next iteration of the conversational cycle the author may revise their text in response to review and feedback.

The second role is that of the critic, in which an argument or proposal from the proponent is examined for understandability, for weaknesses and flaws which are communicated back to the author. Critical appreciation need not be negative and the role of the critic includes that of friendly listener following a line of thinking or a thought experiment sympathetically in order to understand it, and perhaps

offering suggestions for extension or variation for improvement. The critic's role is to report back their responses, reactions or criticisms. This role also encompasses the specialised functions of referee or editor of a mathematical journal. They evaluate a submitted text with respect to community standards in the mathematical specialism, and write and communicate evaluative feedback and make an acceptance decision (in an extended, cultural conversation).

Dutilh Novaes (2021) refers to these two roles as those of *prover* and *skeptic* in her analysis of the dialogical roots of mathematical proof. She includes referees of work submitted for publication under the title of skeptic, as well as mathematicians, more generally, being addressed or persuaded by proposals made by the prover.

Only in the simplest cases are direct or extended conversations just between two persons. The proponent may be a single mathematician or a group of collaborating researchers. Likewise the critic may be a group of referees overseen by an editor who collates the written feedback reports and comes to the decision whether to accept unconditionally, to request revisions and further refereeing, or to reject unequivocally.

In addition, the locations of conversations, whether within institutions, across institutions, or outside established institutions, as well as the power relations and status of the participants are yet further complicating factors, as Dutilh Novaes (2021) reports. The power, status and roles of interlocutors are decisive contributors to dominance in agonistic conversations. Thus critics, including referees and editors, exercise their power in decisions about accepting mathematicians' submissions. They are gatekeepers, fulfilling a vital epistemological function. Most mathematicians will undertake both the roles of proposer and critic in different conversations. The ethics of mathematical practice require that these roles are not in conflict. The editor's power with respect to the publication outlet they oversee should not transfer to their role as proponent when they submit their own articles for publication, neither in the home journal nor elsewhere. Such ethical restrictions serve the purposes of epistemological hygiene, keeping knowledge acceptance processes and products free from the infections of cronyism, corruption, ulterior motives and external interests (Ernest 2020). In some applied areas such as medicine and agro-biological research, instances of the corruption of the refereeing process by large corporations and financial interests are well documented (Goldacre 2012). Thus conversations not purely epistemological processes; there are always ineliminable ethical and power dimensions. There is always the risk of epistemic injustice in mathematics in which some voices are disregarded (Rittberg et al. 2020).

Conversation is thus not simply an epistemological process. Apart from the issues of power and ethics in conversation, there are also subjective and emotional dimensions woven in. As any teacher knows, the tone and wording of any inputs and verdicts offered to a student or a partner in conversation play a large part in their reception. A harsh tone in a criticism may have a large impact, beyond just the response to the knowledge correction involved. In the extreme case hurt feelings can erode the recipients sense of self-worth and lead to the abandonment of a potentially productive development or line of thought. This too is one of the possible outcomes of conversations, especially in educational contexts.

Each mathematician adopts both the positions of proponent and critic. In considering a general claim or law, the mathematician will often seek a counterexample as a means to 'prove' (i.e., test) the law (Lockwood et al. 2016). This is a heuristic learned early on in a mathematician's development. At all levels the research mathematicians will alternate between the roles of proponent and critic, sometimes switching momentarily. Finding a flaw in an argument is not just a means of rejecting it, as it is claimed to be in the empirical sciences (Popper 1959). It is finding a weak point that needs to be bolstered up, to sustain the overall proof, as Lakatos (1976) illustrates.

Conversation between mathematicians is the central mechanism by means of which mathematical knowledge is constructed. Drawing on their knowledge and imagination (capacities already socially shaped) mathematicians construct new proposals, subjecting them to their own inner critic. From this process flows the creation of new, extended or revised mathematical concepts, representations, conjectures, arguments and proofs. Conversation in this context of discovery may result multiple voices with multiple opinions, if there is disagreement about a new mathematical proposal. Here the proposer can ignore some critical responses as she continues to develop the proposal. However, in the context of justification, conversation is the central mechanism through which the testing, critical scrutiny, evaluation and warranting of mathematical knowledge productions takes place. This process determines the acceptance of such proposals, which are then added to the canon of mathematical knowledge via recognised and approved publication outlets. Criticism or rejection cannot simply be ignored if the aim is to publish the proposal in the chosen outlet.

How are disagreements handled? Ramsbotham (2013) argues that in the area of political conflict, engagement and resolution the dominant theoretical perspectives fail to adequately acknowledge the possibilities of radical disagreement. However radical disagreement undoubtedly does occur in mathematical practice, as the following case studies show. Where there is radical disagreement over new mathematical knowledge claims the balance of power favours the 'gatekeepers'; editors of journals, conference referees and publishers of authoritative book series. In these cases the decision makers normally rely on detailed analyses and assessments by expert referees. Typically new mathematical results rely on proofs as their rational justifications. Since the role of proof is to persuade, acceptance decisions rest on expert judgements as to whether the proofs are persuasive. Other significant issues may also arise, such as whether the concepts or procedures in a new theory are coherently defined and defensibly valid. Additionally, the community of mathematicians is fundamentally conservative, showing resistance to new mathematics that is not consistent with existing theories, concepts or practices. However, unlike science, where a revolution is needed to replace one theory by another (Kuhn 1970), in mathematics mutually inconsistent theories can sometimes coexist, side-by-side, as distinct subspecialisms.

The ideal situation in which mathematical submissions are judged rationally and purely on their merits does not always obtain, for personal favouritism, enmity and other factors can distort the process. Communities of mathematicians are normally committed to the unhindered advance of the discipline strongly

enough for many such distortions to be ironed out. However, cases of epistemic injustice in mathematics undoubtedly will occur (Rittberg et al. 2020).

The role of a referee or critic in mathematics is both to indicate areas needing improvement in a submission, as well as making a judgement as to whether the submission is currently acceptable. Thus it is both formative and summative. The referees' assessment passed back to the author enables them to try to improve the article for resubmission. Ultimately, it is a journal editor's role to decide on the acceptability of the submission, to make a summative final judgement. From an epistemic perspective, within the context of justification severe judgements are necessary to maintain truth standards in mathematics.

However, within the context of discovery, the aggressive adversary method (Moulton 1983) may not be the best way to advance mathematics, to bring out the best in a proposer. Tone as well as content of criticism are both important. In mathematics, commonly critiques are directed at specific weaknesses in a proof. As Lakatos (1976) points out, there are strategies for defending a critiqued proof such as 'monster-barring' and 'concept-stretching', that modify the theorem or the concepts involved in order to repair the proof.

Overall, the results of warranting conversations in mathematics are rejection, disagreement or acceptance of the proposals, either unqualified or subject to revisions. Disagreements may well spread beyond the original warranting or acceptance conversations, and controversies emerge when these disagreements, including questioning the judgements of referees and editors, spread beyond the core conversation where gate-keeping decisions were made, to wider discussions about acceptance and publication (or rejection) of the proposals. But, what I have shown is that conversation is the channel by means of which mathematical proposals or texts are formed and then considered, negotiated and critiqued. Conversation is the social process by means of which such warranting takes place. (It should not be overlooked that it is also the process underpinning mathematical formation, invention and discovery).

Before moving on to applications, there is the question of the status and purpose of CT. At least three interpretations are possible, with different temporal orientations.

1. **Past.** CT is a descriptive theory applicable to the history of mathematics and social activities in mathematical practice – a non-judgemental tool for describing what actually occurs.
2. **Present.** CT is a philosophical theory categorising the epistemic activities in social construction of mathematical knowledge, in terms of a logic of question and answer.
3. **Future.** CT is a normative/prescriptive methodology proposing the best and most efficient means of socially constructing mathematical knowledge – including heuristic guidance for practitioners.

My intention in proposing CT here and elsewhere (Ernest 1991, 1998) is endorsing 1 and 2 and rejecting 3. CT is a descriptive theory applicable to past

mathematical practices. As such it is an empirical theory that needs to be tested in practice. Second CT is intended as a philosophical theory of knowledge creation within a social constructionist philosophy of mathematics. However I do not believe that a prescriptive methodology of mathematics is desirable or possible. Human ingenuity continues to overcome the strictures of past practices, as the history of infinity illustrates.

3 Four Episodes in the Social Construction of Infinity

3.1 Cantor's Founding Contributions

Georg Cantor (1845–1918) was a brilliant mathematician whose work started in number theory, moved on to Fourier series, where he needed a theory to deal with collections of points, thus founding set theory. From 1863 to 1867 Cantor studied at the University of Berlin where he became friends with Hermann Schwarz and attended lectures by Weierstrass, Kummer and Kronecker, the last being one of Germany's most renowned and influential mathematicians (Dauben 1978). Cantor was appointed to Halle in 1869 and promoted to Extraordinary Professor there in 1872, and that year he defined irrational numbers in terms of convergent sequences. He also began a friendship with Dedekind. In 1873 Cantor proved the rational numbers and the algebraic numbers countable. In 1877 he proved that there was a 1–1 correspondence between the points on the interval [0, 1] and points in p-dimensional space. Cantor was surprised at his own discovery and wrote:-"I see it, but I don't believe it!" Of course this had implications for geometry and the notion of dimension of a space.

A major paper on dimension which Cantor submitted to *Crelle's Journal* in 1877 was treated with suspicion by Kronecker, and only published after Dedekind intervened on Cantor's behalf. Cantor greatly resented Kronecker's opposition to his work and never submitted any further papers to *Crelle's Journal*. Kronecker was a finitist and he objected to the idea of a completed infinity. His near-lifelong opposition was expressed very strongly, with him claiming that Cantor was a scientific charlatan, a renegade, a "corrupter of youth" (Dauben 1979: p. 1).

Thus, early in his career, Cantor was already having to confront the strong opposition of one of the most eminent mathematicians of his day. Transfinite set theory was still in its infancy, untried and probably seen as suspect by most mathematicians who had been conditioned to reject the absolute infinite in mathematics. Consequently, Cantor's new ideas were particularly vulnerable to opposition like Kronecker's, and Cantor resented what he regarded as unfair and premature criticism (Dauben 1978).

Between 1879 and 1884 Cantor published a series of six papers in *Mathematische Annalen* providing an introduction to his novel theory of sets. Klein may have had a major influence in having *Mathematische Annalen* publish them. However there were a number of problems which occurred during these years which proved difficult for Cantor. Although promoted to full professor in 1879 on Heine's recommendation, Cantor had been hoping for a chair at a more prestigious university. His

long standing correspondence with Schwarz ended in 1880 when Schwarz no longer supported his work and opposition to Cantor's ideas continued to grow. In October 1881 his supportive colleague Heine died but when Dedekind was offered the chair at Halle in 1882, at Cantor's behest, he declined it. Cantor took this personally, and their rich mathematical correspondence ceased. At about this time Cantor began another important correspondence with Mittag-Leffler. Soon Cantor was publishing in Mittag-Leffler's journal *Acta Mathematica.* Cantor also continued with the publication of his important series of six papers in *Mathematische Annalen*. The fifth paper in this series *Grundlagen einer allgemeinen Mannigfaltigkeitslehre* was also published as a separate monograph, because Cantor realised that his theory of sets was not finding the acceptance that he had hoped and the *Grundlagen* was designed to reply to the criticisms with its presentation of the transfinite numbers as an extension of the natural numbers. Cantor was well aware of the strength of the opposition to his ideas, writing "I place myself in a certain opposition to views widely held concerning the mathematical infinite" (O'Connor and Robertson 1998).

In May 1884 Cantor had his first recorded attack of depression. It is now accepted that Cantor's mathematical worries and his difficult relationships, including that with Kronecker, were greatly magnified by his depression but were not its cause. In fact, in 1884 he reached out in an attempt at reconciliation, and Kronecker accepted the gesture. However, it is unlikely that this had any impact on Kronecker's opposition to his work on infinite sets.

Further mathematical worries began to trouble Cantor at this time, for he feared that he could not prove the continuum hypothesis. Then in 1885 Mittag-Leffler persuaded Cantor to withdraw one of his papers from *Acta Mathematica* when it had reached the proof stage because he thought it too controversial. Cantor was very hurt, never submitted to *Acta Mathematica* again, and his correspondence with Mittag-Leffler virtually ceased. For a while he almost gave up mathematics in favour of philosophy, fearing that he had lost his last main mathematical ally (Dauban 1979).

Cantor founded The German Mathematical Society in 1890, and despite the bitter antagonism between himself and Kronecker, Cantor invited Kronecker to address the first meeting, although in the event, he was unable to attend. Cantor was also himself involved in controversy in attacking work on infinitesimals by other mathematicians including Thomae, du Bois-Reymond, Stolz and Veronese (Dauben 1979).

His last major papers on set theory appeared in 1895 and 1897, again in *Mathematische Annalen* under Klein's editorship. Cantor hoped to include a proof of the continuum hypothesis but this was not to be. This failure is regarded as another stressor that influenced Cantor negatively. From the mid 1890s until the end of his life Cantor fought against the mental illness of depression. He continued to teach but only intermittently, and his work in mathematics diminished, and he redirected some of his energies to philosophical, literary and mystical matters.

In 1897 Cantor attended the first International Congress of Mathematicians in Zürich, and Hurwitz and Hadamard openly expressed their great admiration of Cantor's work. He continued working on paradoxes of set theory.

In 1900 Hilbert gave a talk at the International Congress of Mathematicians in Paris in which he posed a list of 23 problems that it turned out did much to shape the direction of mathematics for the next century, The first of these problems was

to determine cardinality of the continuum, that is to settle the continuum hypothesis. Although Cantor was not present, his problem, at the head of the distinguished list, proved that he had well and truly arrived. The continuum hypothesis cannot be formulated, let alone addressed, without the use of Cantor's theory.

Cantor did attend the International Congress of Mathematicians at Heidelberg in August 1904. At that conference Julius König gave a lecture in which he 'proved' that Cantor's Continuum Hypothesis was false. Cantor himself attended this lecture and said at the end how grateful he was to have lived to see this question answered. However, according to Dauben (1978), Cantor was humiliated and distressed at his work being called into question by König. It turned out that the 'proof' was wrong and later Ernst Zermelo demonstrated the error in it. (O'Connor and Robertson 2020).

Throughout his lifetime Cantor's work remained controversial with leading mathematicians like Poincare condemning it, while others like Hilbert supporting it in the strongest terms. Hilbert publicly described it as "…the finest product of mathematical genius and one of the supreme achievements of purely intellectual human activity." (O'Connor and Robertson 1998).

Cantor's life illustrates how much mathematical conversations mattered to him. He was a mathematician of great creativity and insight, and when supported by his colleagues and critics his work flowed and flowered. His correspondence with Schwarz, Dedekind, Klein Mittag-Leffler, Hermite and others and was very helpful in helping to shape and express his ideas. With regard to the supportive Cantor-Dedekind correspondence "almost all of it was written at Cantor's initiative and in order to discuss his new ideas while, in most of the correspondence, Dedekind played only the role of a critic." (Ferreiros 1993, p. 20). Although Cantor's correspondence with Mittag-Leffler was over a shorter period of time, it included more than 100 letters, but after the disagreement about publishing Cantor's paper in *Acta Mathematica*, Cantor's mathematical output dried up for a while, demonstrating its impact on him.

Controversy based on negative criticism and personal attacks appear to have hit Cantor very hard, robbing him of his confidence and staunching his flow of creative output. The withdrawal of support by close allies such as Dedekind and Mittag-Leffler left him feeling betrayed, exposed and greatly weakened. We know that his subject matter, infinite arithmetic and set theory, is one of the most controversial in all of mathematics, flying in the face of mathematical tradition. Unfortunately, Cantor was not robust enough to weather the excessive criticism that he received, coupled with the withdrawal of support from mathematical friends, especially as he was prone to the mental illness of depression. A close examination of medical records and other evidence suggests that Cantor was bipolar (Dauben 1978). He leaves behind one of the most outstanding achievements to come from any one person in the history of mathematics. But he also illustrates the impacts of the winds of support and opposition, bolstering him up and blowing him down. His conversations with those around him, as well as those

about him were powerful influences both in the domains of knowledge and that of affect, including emotion, attitudes and confidence.

Cantor's career ran the whole gamut of rejection, disagreement and acceptance, and was awash, throughout, with controversy. There is an analogy between the impact of Cantor's work and one of Kuhn's (1970) scientific revolutions. Cantor's set theory and transfinite number theory constitutes a new paradigm for modern mathematics, and only after his critics had died off did it become almost universally accepted, although a strand of opposition survives in the form of the intuitionism.[1]

A more detailed study of Cantor's correspondence and work would reveal the specific ways that conversation shaped his concepts, theories and proofs. An examination all his records and papers should provide indications of the intrapersonal conversations, Cantor's creative ideas and his self criticisms (or other's) which shaped his mathematical theories and contributions to knowledge. However, this would require an in-depth study way beyond the scope of this paper. Here I focus primarily on the major documented social processes of rejection, disagreement and acceptance, milestones in his career and contributions.

3.2 The Intuitionists' and Constructivists' Rejection of Completed Infinities

The late 19th century saw a number of antinomies and contradictions in mathematics, especially following the development of set theory and infinite sets, introduced by Cantor. This was widely viewed as a crisis in its foundations and three main schools in the philosophy of mathematics emerged to safeguard mathematics. Logicism sought to ground mathematics in logic and proof, and formalism sought to safeguard mathematics through a finitist metatheory, with mathematical theories themselves viewed a formal syntactical 'games'. However, third strand, that of intuitionism, followed a programme of cutting mathematics back, disallowing completed infinities and non-constructive modes of reasoning.

Although there are antecedents in finitistic thought, in such as with Kronecker and Poincare, it was L. E. J. Brouwer who founded intuitionism. One of his basic principles is that: "to exist in mathematics means: to be constructed by intuition" (Brouwer 1907, p. 177). Construction is a step by step process that we can imagine, and in following such steps we can never arrive at a completed infinity. So its existence is ruled out.

Brouwer's intuitionism was developed in his 1907 PhD thesis, where the philosophical basis of his approach was first stated. Brouwer's dissertation consisted of three chapters: 'The construction of mathematics,' 'Mathematics and experience,' and 'Mathematics and logic'. During its development his distinguished supervisor Korteweg's main criticism was directed towards the second chapter. This focussed on philosophical issues and drew on the ideas in Brouwer's (1905) booklet *Life, Art, and Mysticism*. Korteweg was a strong supporter of Brouwer's emerging

[1] An important part of the resistance to Cantor's ideas is that he introduced formalism, in the sense that the formal structure of mathematics defines its possible meanings, rather than a designated model pre-assigning mathematical meanings to the terms of a theory. This is why Hilbert found his mathematics so sympathetic whereas Poincare did not.

intuitionism, first expressed in his thesis, but felt that this philosophy and mystical tract was inappropriate for a scientific dissertation. In response, Brouwer revised this second part of his thesis to cut back its philosophical and mystical contents (Bar-On 2020). Although it is claimed that the 1905 booklet "contains his basic ideas on mind, language, ontology and epistemology" (Atten 2020), it does not mention mathematics.

Right from the outset, Brouwer's intuitionism was radically counterpoised against classical mathematics. Brouwer rejected completed infinities, the law of the excluded middle (except in finite cases) and indirect or negative existence proofs. The dispute has never ended, but was at its peak in the 1920 and 1930 s. Intuitionism has two principal theses, one positive and one negative. The positive thesis is that constructive arguments and proofs are uniquely valuable and should be sought above all other forms of reasoning. The negative thesis is that only constructive arguments and proof s have any value or meaning, and any that are non-constructive (employing completed infinities, the full law of the excluded middle or negative existence proofs, etc.) have neither meaning nor value and should be expunged from mathematics. This second principle delegitimises a great deal of mathematics and unsurprisingly created hostile replies from many mathematicians.

> At issue in the sometimes bitter disputes was the relation of mathematics to logic, as well as fundamental questions of methodology, such as how quantifiers were to be construed, to what extent, if at all, nonconstructive methods were justified, and whether there were important connections to be made between syntactic and semantic notions. (Dawson 1997, p. 1).

For example, when Fraenkel (1927) published an account of Brower's intuitionism, The relationship which started in friendship and perfect harmony, ended in irritation, reproaches, and even hostility. At the same time "Hilbert's attacks got more personal than ever, and the debate had turned from a scholarly exchange into a conflict that threatened to split the German mathematical community." (Dalen 2000, p. 287).

Over the course of the decades-long debate, Hilbert had lost his 'gifted pupil' Herman Weyl to intuitionism, for some years. A symposium on the foundations of mathematics appeared in Erkenntnis in 1931 with elaborations of the positions of the three schools of logicism, formalism, and intuitionism. The last of these three was defended by Brouwer's student Heyting, who took up the mantle alongside his mentor.

Heyting's (1956) interpretation of intuitionism is presented in dialogical form, with six disputants including a classical mathematician, formalist, intuitionist, pragmatist, and two others concerned with letters and signs. The dialogue moves from preliminary philosophical matters to an in-depth consideration of mathematical topics from an intuitionist slant, including arithmetic, spreads and species (constructive versions of real numbers, sequences and sets), algebra, plane topology, measure and integration, logic and other controversial subjects (sic.). Thus, what is offered is a serious alternative to classical mathematics, not just philosophically, but also in terms of mathematical content, methods and proofs. Although Heyting made

significant contributions himself, the book is primarily based on Brouwer's work and includes three pages of references to the latter's publications.

Of particular importance in the book is the interpretation of the logical connectives and quantifiers, developed by Heyting and Kolmogorov, and inspired by Brouwer. This interpretation has proof as its central notion, rather than the truth found in the standard semantics for classical logic. For example:

- a proof of ¬A is a derivation of a contradiction from the assumption of A.
- a proof of A → B is a construction that transforms any proof of A into a proof of B.
- a proof of ∀xA(x) is a construction that for any given object a yields a proof of A(a).

The rules of proof are such that if one begins with finite constructive sentences of mathematics the results must also be of this type.

Franchella (2019) shows how Brouwer and Heyting maintained their commitment to and promotion of intuitionism throughout their lives, using the three terms logicism, formalism, intuitionism as a way of labelling their philosophical opponents, despite the changing meanings of these schools of thought. Bar-On (2020) shows how three enduring demands of the intuitionist programme elicited major opposition throughout. These were Brouwer's idiosyncratic philosophical agenda to reform mathematics; the changes required on the whole body of mathematical knowledge, with the resultant exclusion of major theories and proofs, and the introduction of new unfamiliar and alien concepts and theories to replace the non-constructive ones. This disagreement and controversy always accompanied intuitionism, in particular, because of its rejection of modern set theory and completed infinities. A cadre of supporters lives on. For example, Dummett (1977) extended the philosophical and logical foundations of intuitionism. Within the subject itself, a tranche of constructive mathematicians emerged from the 1930s onwards, carrying forward a stripped down version of the positive thesis of intuitionism. These included Stephen Kleene, Errett Bishop and Andreĭ A. Markov, known as the founder of the Soviet school of constructive mathematics (Vandoulakis 2015).

Bishop (1967) showed, most impressively, that it is possible to provide constructive foundations for analysis, developing Thesis (1) But Bishop's agenda is not merely to promote constructive mathematics. In his *Schizophrenia in Contemporary Mathematics* (Bishop 1973) attributes a list of what he terms schizophrenic attributes to contemporary mathematics including rejection of common sense in favour of formalism; debasement of meaning by wilful refusal to accommodate certain aspects of reality; and the (in)appropriateness of the means adopted to achieve the ends of mathematics, thus advancing Thesis (2) While there is no mention of infinity in this booklet, it is clear that he strenuously rejects the concept of completed infinities. The controversy lives on. The contents of Bishop (1973) were presented as an extended lecture to American Mathematical Society "The reception of the audience became tinged with active anger." (Greenleaf 2020, p. 14). Greenleaf himself continues the fight for Thesis 2, arguing that addiction is a better term than schizophrenia,

as classical mathematicians refuse to give up completed infinities and the other non-constructive concepts, methods and results of classical mathematics.

Intuitionism and constructivism demonstrate that there can be enduring disagreements and controversies within mathematics. Traditional mathematicians maintain that constructive mathematics is part of classical mathematics (Tait 1983). Indeed, all of intuitionist and constructive mathematics can be translated into classical mathematics in a truth preserving way. But constructivists protest that such translations radically distort the meanings and goals of the mathematics into something unacceptable to themselves. These disagreements are unlikely to go away while representatives of both sides survive.

From the point of view of conversation, it is clear that there has been an extensive and spirited interpersonal dialogue. The enduring dispute has clarified and benefitted the claims of both sides in the argument. The underlying assumptions and principles of both sides have been made clear, as have the differences between opposing sides. There is no doubt that intuitionism and constructivism have enriched mathematics, with the acceptance of constructive mathematics within the pale of classical mathematics. Atten (2014) even suggests that Brouwer's ideas prompted Gödel to go in the direction that led to his celebrated incompleteness theorems. Indeed Gödel (1931) was very careful to stick to primitive recursive functions throughout his proof and make the whole enterprise intuitionistically acceptable, as he explicitly says.

Brouwer's presentation of intuitionism as based on subjective intuition invites an intrapersonal conversational analysis, although he foregrounds the role of the proponent with its intuition of time (before/after generating 'twoity') and the intuitive creation of new mathematical entities, but downplays the role of critic (Iemhoff 2020). Brouwer himself would not welcome this approach as he regards mathematics as a languageless creation of the mind, and thus rejects the CT view that language is an internalized part of intrapersonal conversation.

What this brief case study shows, and the role of infinity illustrates, is that there can be legitimate disputes and disagreements within mathematics that are irresolvable. Completed infinities are welcomed, accepted and studied on one side, and wholly rejected and delegitimised on the other. In this case mathematics resembles philosophy, where some problems and disputes are never overcome, being based on incompatible conceptual frameworks used to address intractable philosophical problems. However, in mathematics itself, as opposed to the philosophy of mathematics, such disagreements are much rarer.

3.3 Sergeyev's Introduction of the Infinite Element 'Grossone'

A contemporary controversy concerns Sergeyev's introduction of the infinite element 'Grossone'. The term 'Grosse' was used by Gauss to refer to a completed infinity and Sergeyev has introduced the terms Grossone, Grosstwo, Grossthree, …, as names for successive completed infinities.

Sergeyev extends the set of natural numbers N with the addition of an infinite element ① (Grossone). He defines the number operations as conservative extensions of number operations on Q, but drawing on those on Euclid's Common Notion no.

5 "The whole is greater than the part" when it comes both to ordering and division. His basic definitions are as follows.

> **The Infinite Unit Axiom.** The infinite unit of measure is introduced as the number of elements of the set, *N*, of natural numbers. It is expressed by the numeral ① called *grossone* and has the following properties:
> *Infinity.* Any finite natural number n is less than grossone, i.e. $n < ①$.
> *Identity.* The following relations link ① to identity elements 0 and 1
> $0\times① = ①\times 0 = 0$; $① - ① = 0$;
> $①/① = 1$; $①^0 = 1$; $1^① = 1$; $0^① = 0$
> *Divisibility.* $①/n$ results in n sets $N_{k;n}$ of the form $\{k, k+n, k+2n, k+3n, \ldots\}$ For any finite natural number k such that $1 \le k \le n$. $N_{k;n}$ is the nth part of the set, *N*, of natural numbers. Each of the $N_{k;n}$ have the same number of elements indicated by the numeral $①_n$ and the union of all n parts is *N*.
> (Sergeyev 2017: p. 236)

Division thus partitions *N* into disjoint subsets. Operations on all the Natural numbers plus Grossone result in an enlarged set called the Extended Natural Numbers (ENN). A generalised member of the set of ENN is a polynomial made up of different powers of Grossone with units in the form $k\times①^0$, comprising infinite number parts in the form of $j\times①^n$ and infinitesimal parts in the form $m\times①^{-n}$, for all natural numbers $n > 0$.

Sergeyev has created a computational device called the Infinity Computer which calculates with these ENN. He has published scores, if not hundreds, of papers on this approach in standard international journals, especially on applications of his transfinite calculational approach, since Sergeyev (2003, 2008) or earlier. His emphasis is on providing a novel means of computation, with a wide variety of applications, as opposed a new foundational theory. Lolli (2015) has attempted to put the theory of Grossone on a sound axiomatic footing, working from a foundational perspective. Sergeyev's applications include work by himself and colleagues in computational optimization and applications, solving ordinary differential equations, probability, game theory, cellular automata, chaos and fractals, and other fields.[2] Sergeyev's (2003) book length treatment of Grossone methods was well reviewed by Pardalos (2006).

However, disagreement and controversy arose because strong criticism has been directed at Sergeyev's Grossone theory since the publication by Gutman and Kutateladze (2008), if not earlier.

Further criticism was elicited when Sergeyev (2015a) published an account of an application of his approach to Olympic medal ranking scores of different countries. An immediate critical response came from Shen (2015) whose judgement, with little supporting argument apart from citing a book review, was reflected in the title of his letter, namely "nonsense". The Editor-in-Chief of The Mathematical Intelligencer invited Sergeyev to respond to Shen's letter which he did immediately, on the following two pages of the journal (Sergeyev 2015b). Here

[2] Sergeyev's website at https://www.theinfinitycomputer.com/ lists a broad range of topics addressed by the publications related to Grossone, from computational optimization to chaos and fractals.

he briefly reiterates his method, its areas of applicability, and provides references. However, since Olympic medal ranking is based on alphabetic ordering, which requires only finitistic methods, this is far from a decisive demonstration of the power of Grossone methods.

Gutman, Katz, Kudryk and Kutateladze wrote to the Editor-in-Chief offering an article in rebuttal entitled "The Mathematical Intelligencer Flunks the Olympics" with the opening sentence "The Mathematical Intelligencer recently published a note by Y. Sergeyev that challenges both mathematics and intelligence." This polemical paper in "five successive versions" was rejected and appeared elsewhere. (Gutman et al. 2017, p. 553)

That year Sergeyev (2017) published very extensive survey paper, 102 pages long, in *European Mathematical Society Surveys in Mathematical Sciences.* This led to a further big controversy. The paper was published online on November 13. Although long, it is not the longest paper the journal has published. The dating on the paper "Received 14 July, 2017; revised 25 August, 2017" suggests that the refereeing was done rapidly. It is claimed to have been properly refereed, with one referee making over 100 comments (private communication from Sergeyev). Later, editorial board confirmed that the acceptance procedures were in no way deficient.

One critic, Alexander Shen, wrote to the editors on November 29, expressing his concern in the strongest possible terms. "the reputation damage to the Journal, to EMS and to mathematical community as a whole cannot be fully undone, but a clear apology/retraction seems to be needed." (Stern 2017, p. 1).

In December 2017, the editorial board of EMS Surveys in Mathematical Sciences issued a statement about the paper, saying its acceptance was 'a serious mistake' and "decision to publish a controversial paper…was apparently made without consulting the editorial board.". The two editors-in-chief of the journal 'assumed responsibility for these mistakes' and resigned from their positions. However, a spokesperson for the journal reported that the paper is not likely to be retracted (Stern 2017).

The same source reports communications with Sergeyev.

> Sergeyev told us that, for the past decade, his work has been under "violent attack" by a group of mathematicians and calls the most recent actions of the journal's editorial board a "witch-hunt campaign:"
> *"I regret to see that the EMS is involved in this campaign that persecutes a person just because he thinks differently and works using new tools."* (Stern 2017)

This source closes its report of the affair with a final word to the critics.

> However, Shen and two mathematicians from Novosibirsk State University in Russia—S. Kutateladze and Alexander Gutman—have published critiques of Sergeyev's work. Gutman told us that he thinks the current paper, along with other articles by Sergeyev, are disrespectful to modern mathematics:
> …

Kutateladze added:
*"The recent publication of the paper by Sergeyev in European Mathematical Society Surveys was a scandalous blunder." (*Stern 2017)

The strength of disagreements makes this a fascinating case study into the reception of new ideas and methods in mathematics.

History will decide the outcome of this controversy, but the number of supporters of Sergeyev continues to grow. For almost two decades he, together with collaborators. followers and students, has published hundreds of papers and books in respectable looking journals, several with good Scopus ratings and citation indices in the Web of Science. Sergeyev has given scores of plenary lectures worldwide and is the recipient of a growing list of prizes and honours. Above all, Sergeyev has made his methods and results public, open to evaluation and scrutiny. While I don't believe his methods constitute a theoretical breakthrough anywhere near the order of Cantor or Robinson (1966), they are claimed by supporters to have both value and utility.

As one reviewer of the long EMS paper puts it, in a mixed review: "Sergeyev has forged an original path to these infinities, infinitesimals and their calculus. He shows how to use his ideas by many examples. However, he has not attempted to write a logical foundation for his ideas." (Kauffman 2021).

This example illustrates the role of conversation in mathematics at several levels. First there is Sergeyev communicating with colleagues face to face in small spaces and national and international lectures, including responses to questions, requests for clarification and criticisms. Second, there are the written communications of his work describing the ideas and results and directed at journals and some published books. At this level there are also the critical replies from referees and the author's own responses included corrected submissions. Then there are meta-level communications directed at journal editors and critics (not to mention those directed at Sergeyev himself) concerning private and public responses to his work, as opposed to in-house journal communications. At this level we find strongly negative publicly made critical responses, as well as sympathetic if critical analyses, and statements containing positive evaluations.

In this controversy, the role of the content cannot be ignored. Infinities and their inverse cousins infinitesimals have always been 'hot potatoes', eliciting strong reactions from those that accept them and those that don't. Such controversies also arise between those who believe these ideas can only be treated in certain prescribed and delimited ways, and others with the courage or foolhardiness to transgress these limits. However, the jury is still out on the value if not the validity of the Grossone approach. No decisive problem has been solved by it as yet that cannot be solved by already existent methods.

3.4 Saburou Saitoh's Controversial 'Division by Zero Calculus'

Saburou Saitoh is a Japanese mathematician known for his theory of reproducing kernels and its applications and other contributions to analysis (Saitoh 1988; Saitoh and Sawano 2016) and whose achievements are celebrated in the festschrift (Matsuura 2015). For the past decade plus he has been developing his 'Division

by Zero Calculus' in a series of publications including Saitoh (2021a) and a number of web-published papers.[3]

The foundational assumption is that the rejection of Brahmagupta's definition of $0/0 = 0$ (zero divided by zero equals zero) is erroneous, and has been wrong for over 1300 years.

Thus a central claim is that

> The division by zero is uniquely and reasonably determined as $1/0 = 0/0 = z/0 = 0$ (Saitoh 2021b: pp. 32–33).

The explicit injunction of Sen and Agarwal (2016, p. 11) that "'Thou shalt not divide by zero' remains valid eternally" is referred to as a common misunderstanding.

In general Saitoh rejects the classical idea that $1/0 = \infty$. (Saitoh 2021b: p. 75). He argues, however, that in some contexts

> $1/0 = \infty$. (15.1). $1/0 = 0$. (15.2) …
> Note that both identities (15.1) and (15.2) are right in their senses. Depending on the interpretations of 1/0, we obtain INFINITY and ZERO, respectively. (Saitoh 2021b: p. 303-4).

The way that the arguments proceed is to interpret 0, division by 0, or a limit approaching zero idiosyncratically within the specific context of certain mathematical problems. Thus, for example, in Kuroda et al. (2014), new meanings of the division by zero and new interpretations of $100/0 = 0$ and $0/0 = 0$ are provided.

There are mathematical structures in which $a/0$ is defined for some a such as in the Riemann sphere (a model of the extended complex plane) and the projectively extended real line, and Saitoh does use these examples. However, such structures do not satisfy every ordinary rule of arithmetic (the field axioms).

Beyond the immediate rejection of the above contradictory assertions, the standard mathematical responses to these assertion are that 0/0 and n/0 are left undefined. Division by the additive identity is simply undefined in a Field, an algebraic structure exemplified by the rational numbers Q, the real numbers R and the complex numbers C. Furthermore infinity ∞ is not a member of any of the standard sets of arithmetical numbers N, Z, Q, R, or C.

Division by 0 is left undefined, because this operation cannot satisfy the definition of division, nor can it remain consistent with other properties of arithmetic. For example, if we are doing arithmetic on members of a set that is not a field, division of a by b is the unique number c satisfying $a = c \times b$. If $b = 0$, then there is no number c such that $a = c \times 0$, if $a \neq 0$. But if $a = 0$, then *every* number c satisfies $0 = c \times 0$.

If we are working in a field and try to invent a unique multiplicative inverse 0^{-1} of 0, we lose associativity of multiplication, because $0^{-1} \times (0 \times a) = 0^{-1} \times 0 = 1$, but

[3] The website https://vixra.org/author/saburou_saitoh accessed 11 June 2021 offers around 40 papers by Professor Saitoh. A new International Journal of Division by Zero Calculus is announced at https://romanpub.com/dbzc-vol-1--2021.php (accessed 11 June 2021).

with associativity then also $(0^{-1} \times 0) \times a = 1 \times a = a$. So you either sacrifice associativity or consistency.

Thus the claims of Saitoh to have invented a novel 'Division by Zero Calculus' are overblown, and in the context of arithmetic, trivially incorrect. To assert $1/0 = \infty$ is to commit two sins, neither ∞ nor division by zero are legitimate within standard arithmetic. To also assert $1/0 = 0$ is to create a contradiction in arithmetic, which renders the whole system meaningless and invalid.

From the perspective of controversy, have there been widespread criticism or protests? Have any voices been raised in protest in these seemingly illegitimate concepts and claims? The answer is in the negative. These publications have simply been ignored. There has been no detectable interest in the claims, and detailed web searches have thrown up no disagreements or rejections. If mathematicians have happened upon these publications and claims they have simply ignored them. My assumption is that if and when this has come to the attention of mathematicians they have been uninterested and not regarded it as worth their attention. So the wider conversation is almost non-existent. Perhaps being ignored is an even worse fate for a mathematician than being the centre of a controversy.

How have Saitoh's many works on 'Division by Zero Calculus' been published? The answer is that they have been self published. Those that have not simply been uploaded to the web, such as Kuroda et al. (2014) and Saitoh (2021a) are published by companies included in Beall's List (2021) of potential predatory publishers. These are condemned because authors simply pay to publish with them without any proper refereeing practices.

To be heard your voice must be part of a conversation with other participants. There was a time, as Cantor's history shows, when there were a small number of central journals that were attended to by most serious mathematicians. While there are many more journals in mathematics these days, mathematicians and other scholars attend to known journals and respectable publishers and ignore the growing numbers of dubious publication sources. So perhaps the neglect of Saitoh by the academic community of mathematicians is not surprising. You have to be in the conversation to be part of the conversation.

4 Conclusion

This paper has explored the reception of contributions concerning or involving infinity in four short case studies using CT. The case of Saitoh shows what can happen if the conversation never spreads beyond a small circle close to the author. Such a restricted conversation can conceivably be helpful in the generation of new ideas. Whether this restricted conversation has yielded any serious critiques, such as in the preceding paragraphs, is unknown in the case of Saitoh. However, without submission to a formal institutional conversation that warrants results the proposals will not be subjected to rigorous critique, and will normally not be accepted as a contribution to the body of mathematical knowledge.

As I have shown, both the roles of the proponent and that of the critic are necessary in the production and warranting of knowledge. In particular, in the context of

justification the critic plays a central role in gatekeeping; maintaining quality control in the production of knowledge. However, this does introduce the risk of epistemic injustice (Rittberg et al.). The case studies illustrate that pressure can be brought to bear to sway acceptance judgements, both positively and negatively, as the example of Cantor and the Sergeyev reveal.

Is CT helpful in understand rejection, disagreement, and acceptance of mathematical concepts and results in mathematical practice? CT describes the mechanism of social interaction in which proponents of new ideas and theories are met with varieties of critical responses. CT accommodates the dialectical patterns of proposal and critical response. This is part of an extended dialogue that can shape both the opinions of the interlocutors and well as the would-be knowledge item proposed.

The account presented here is limited, having focussed most on the social context of justification; the arena in which controversy, rejection, disagreement and acceptance all take place. A deeper, historical analysis of the conversational interactions involved would strengthen the case for CT, but the level of detail involved would require book length treatment. Beyond this, studies of the role of CT in the context of discovery, showing how concepts and proofs are originated, critiqued and modified, such as Lakatos (1962) sketches, would offer a powerful endorsement of the broader value of CT.

This paper explores the range of interactions of proponents and critics, and the roles adopted within the controversial history of infinity. Rejection, disagreement and acceptance have been accommodated within the descriptive framework of CT, presented as a tool for understanding the social mechanisms of knowledge generation and validation. As such it offers greater verisimilitude than the traditional logical reconstruction of concepts and proofs that smooths out the actual disagreements, detours and redefinitions that occurred in the history of mathematics and which led to our modern theories (Lakatos 1962).

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